

- Many experience and/or can be significant in amount

Sedation



- Many experience and/or can be significant in amount
- Sedation is usually transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia

Best Augmenting Agents for Side Effects

- * Augmentation of thioridazine has not been systematically studied and can be dangerous

Dosing and use

Usual Dosage Range

- 200–800 mg/day in divided doses

Dosage Forms

- Tablet 10 mg, 15 mg, 25 mg, 50 mg, 100 mg
- Liquid 30 mg/mL, 100 mg/mL (discontinued in USA)
- Suspension 5 mg/mL, 20 mg/mL (discontinued in USA)

How to Dose

- 50–100 mg 3 times a day; increase gradually; maximum 800 mg/day in divided doses



Dosing Tips

- * Prolongation of the QTc interval is dose-dependent, so start low and go slow while carefully monitoring QTc interval
- Pigmentary retinopathy has been reported in patients taking doses exceeding the recommended range
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Sedation, confusion, respiratory depression, cardiac disturbance, hypotension, seizure, coma

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)

- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after thioridazine is discontinued

Pharmacokinetics

- Metabolized by CYP450 2D6



Drug Interactions

- May decrease the effects of levodopa, dopamine agonists
- May increase the effects of antihypertensive drugs
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- * CYP450 2D6 inhibitors including paroxetine, fluoxetine, duloxetine, bupropion, sertraline, citalopram, and others can raise thioridazine to dangerous levels
- * Fluvoxamine, propranolol, and pindolol also inhibit thioridazine metabolism and can raise thioridazine to dangerous levels
- Respiratory depression/arrest may occur if used with a barbiturate
- Additive effects may occur if used with CNS depressants
- Alcohol and diuretics may increase the risk of hypotension; epinephrine may lower blood pressure

- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome

Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- * Thioridazine can increase the QTc interval and potentially cause torsade de pointes-type arrhythmia or sudden death, especially in combination with drugs that raise its levels
- Use cautiously in patients with respiratory disorders, glaucoma, or urinary retention
- Avoid extreme heat exposure
- Antiemetic effect can mask signs of other disorders or overdose
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Do not use epinephrine in event of overdose, as interaction with some pressor agents may lower blood pressure
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Observe for signs of pigmentary retinopathy, especially at higher doses

- Because thioridazine may dose-dependently prolong QTc interval, use with caution in patients who have bradycardia or who are taking drugs that can induce bradycardia (e.g., beta blockers, calcium channel blockers, clonidine, digitalis)
- Because thioridazine may dose-dependently prolong QTc interval, use with caution in patients who have hypokalemia and/or hypomagnesemia or who are taking drugs that can induce hypokalemia and/or magnesemia (e.g., diuretics, stimulant laxatives, intravenous amphotericin B, glucocorticoids, tetracosactide)
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression
- If patient suffers from extreme hypertension/hypotension
- * If QTc interval greater than 450 msec or if taking an agent capable of significantly prolonging QTc interval (e.g., pimozide, selected antiarrhythmics, moxifloxacin, and sparfloxacin)

- * If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- * If patient is taking drugs that inhibit thioridazine metabolism, including CYP450 inhibitors
- * If there is reduced CYP450 2D6 function, such as in patients who are 2D6 poor metabolizers
- If there is a proven allergy to thioridazine
- If there is a known sensitivity to any phenothiazine

Special populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Thioridazine produces a dose-dependent prolongation of QTc interval, which may be enhanced by the existence of bradycardia, hypokalemia, congenital or acquired long QTc interval, which should be evaluated prior to administering thioridazine
- Use with caution if treating concomitantly with a medication likely to produce prolonged bradycardia, hypokalemia slowing of

intracardiac conduction, or prolongation of the QTc interval

- Avoid thioridazine in patients with a known history of QTc prolongation, recent acute myocardial infarction, and uncompensated heart failure
- * Risk/benefit ratio may not justify use in cardiac impairment

Elderly

- Some patients may tolerate lower doses better
- Elderly patients may be more sensitive to adverse effects, including agranulocytosis and leukopenia
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events



Children and Adolescents

- Safety and efficacy not established in children under age 2
- Dose: initial 0.5 mg/kg per day in divided doses; increase gradually; maximum 3 mg/kg per day
- Risk/benefit ratio may not justify use in children or adolescents

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotic may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy
- Should evaluate for an antipsychotic with a better risk/benefit ratio if treatment is required during pregnancy

Breast Feeding

- Unknown if thioridazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The art of psychopharmacology

Potential Advantages

- Only for patients who respond to this agent and not other antipsychotics

Potential Disadvantages

- Children
- Elderly
- Patients on other drugs
- Those with low CYP450 2D6 metabolism

Primary Target Symptoms

- Positive symptoms of psychosis in patients who fail to respond to treatment with other antipsychotics
- Motor and autonomic hyperactivity in patients who fail to respond to treatment with other antipsychotics

- Violent or aggressive behavior in patients who fail to respond to treatment with other antipsychotics

Pearls

- * Generally, the benefits of thioridazine do not outweigh its risks for most patients
- * Because of its effects on the QTc interval, thioridazine is not intended for use unless other options (at least 2 antipsychotics) have failed
- Thioridazine has not been systematically studied in treatment-refractory schizophrenia
- * Phenotypic testing may be necessary in order to detect the 7% of the Caucasian population for whom thioridazine is contraindicated due to a genetic variant leading to reduced activity of CYP450 2D6
- Conventional antipsychotics are much less expensive than atypical antipsychotics
- Thioridazine causes less drug-induced parkinsonism than some other conventional antipsychotics
- * Was once a preferred antipsychotic for children and the elderly, and for those whose symptoms benefited from a sedating low-potency phenothiazine with a lower incidence of drug-induced parkinsonism
- * However, now it is recognized that the dangers of cardiac arrhythmias and drug interactions outweigh the benefits of

thioridazine, and it is now considered a second-line treatment if it is considered at all

Suggested Reading

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Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Leucht S , Wahlbeck K , Hamann J , Kissling W . New generation antipsychotics versus low-potency conventional antipsychotics: a systematic review and meta-analysis . *Lancet* 2003 ;**361** :1581 –9.

Thiothixene

Therapeutics

Brands

- Navane
- *see index for additional brand names*

Generic?

Yes

Class

- Conventional antipsychotic (neuroleptic, thioxanthene, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

- * Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes

(fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and thiothixene should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia

- * Drug-induced parkinsonism
- * Tardive dyskinesia
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea
- Sedation
- Dry mouth, constipation, vision disturbance, urinary retention
- Hypotension, tachycardia
- Rare fine lenticular pigmentation



Life-Threatening or Dangerous Side Effects

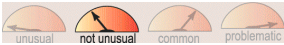
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- Rare blood dyscrasias
- Rare hepatic toxicity
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- For sedation, take at night
- Reduce the dose
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 15–30 mg/day

Dosage Forms

- Capsule 2 mg, 5 mg, 10 mg

How To Dose

- Initial 5–10 mg/day; maximum dose generally 60 mg/day; higher doses may be given in divided doses



Dosing Tips

- When thiothixene is dosed too high, it can induce or worsen negative symptoms of schizophrenia
- Lower doses may provide the best benefit with fewest side effects in patients who respond to low doses

- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Muscle twitching, sedation, dizziness, CNS depression, rigidity, weakness, torticollis, dysphagia, hypotension, coma

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after thiothixene is discontinued

Pharmacokinetics

- Initial elimination half-life approximately 3.4 hours
- Terminal elimination half-life approximately 34 hours



Drug Interactions

- Respiratory depression may occur when thiothixene is combined with lorazepam
- Additive effects may occur if used with CNS depressants
- May decrease the effects of levodopa, dopamine agonists
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Combined use with epinephrine may lower blood pressure
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions thiothixene may

antagonize

Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Antiemetic effect can mask signs of other disorders or overdose
- Do not use epinephrine in event of overdose, as interaction with some pressor agents may lower blood pressure
- Use cautiously in patients with glaucoma, urinary retention
- Observe for signs of ocular toxicity (pigmentary retinopathy, lenticular pigmentation)
- Avoid extreme heat exposure
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient has CNS depression, is in a comatose state, has circulatory collapse, or there is presence of blood dyscrasias
- If there is a proven allergy to thiothixene

Special populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Thiothixene may cause or aggravate ECG changes
- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events



Children and Adolescents

- Safety and efficacy have not been established in children under age 12
- Generally consider second-line after atypical antipsychotics



Pregnancy

- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy